

MA511: Computer Programming

LAB 3 Assignment (Due: 8th Aug 2025)

To execute these labs, you will only use the tools/techniques we have discussed in class.

1. Write a C program, that prompts the user to enter a positive integer, then repeatedly shows a menu and does the corresponding task:
 - 1) Check if the input number is prime.
 - 2) Check if the input number is an Armstrong number. Print Yes or No.
 - 3) Exit

A number is called an Armstrong number if the sum of the cubes of the digits of the number is equal to the number. For example, $153 = 1^3 + 5^3 + 3^3$.

2. Write a C program that reads a single character from the user to identify if the character is a vowel (a, e, i, o, u), a digit (0-9), or something else. The program should continuously run until the user inputs an EOF.
3. Write a C program to find the HCF of two numbers.
4. Write a C program to accept the basic salary of an employee from the user. Calculate the gross salary on the following basis:

Basic	HRA	DA
1 - 4000	10%	50%
4001 - 8000	20%	60%
8001 - 12000	25%	70%
12000 and above	30%	80%

For example, if the user inputs 4500 as basic, the gross salary is

$$4500 + .20 * 4500 + .60 * 4500$$

---000---